

Splice Locations

- Naive guess: at $\frac{1}{4}$ and $\frac{3}{4}$ of bridge
 - Good balance of moment and shear
- Code
 - Find FOS due to select important cases both for shear and moment, and splice location is at when they intersect
 - Cases for shear:
 - **Can cause glue on splices to shear off**
 - Cases for moment:
 - **Flexural stress --> Tension at bottom can separate glued sections**
 - Compression on top --> buckling/crushing --> not super relevant
 - Sides
 - Buckling
 - Not relevant since geometry dependent
 - Second order efforts (shear due to buckling deformation) are not considered due to lack of theory
 - Tension
 - Tension at maximum at bottom, so same as the bottom splice case
 - Naive guess for multiple layers
 - Layer thickness is total thickness
 - Strength is weighted average of glued surface strength and matboard strength
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